

Numerical Analysis homework V

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I. Detailed proof of Thm

We first check that the operations that $\mathcal{C}[a, b]$ is closed under them.

Closure under addition. Let $f, g \in \mathcal{C}[a, b]$. For each $x \in [a, b]$ we define $(f + g)(x) := f(x) + g(x)$. Since f and g are continuous and the sum of continuous functions is continuous, the function $f + g$ is continuous on $[a, b]$. Hence $f + g \in \mathcal{C}[a, b]$.

Closure under scalar multiplication. Let $f \in \mathcal{C}[a, b]$ and $\alpha \in \mathbb{C}$. For each $x \in [a, b]$ define $(\alpha f)(x) := \alpha f(x)$. The map $x \mapsto f(x)$ is continuous, and multiplication by the constant α is a continuous operation, so the product αf is continuous on $[a, b]$. Thus $\alpha f \in \mathcal{C}[a, b]$.

Now we verify the vector space axioms. All equalities below are understood pointwise on $[a, b]$.

- *Associativity of addition:* For any $f, g, h \in \mathcal{C}[a, b]$,

$$((f + g) + h)(x) = (f(x) + g(x)) + h(x) = f(x) + (g(x) + h(x)) = (f + (g + h))(x).$$

- *Commutativity of addition:* For any $f, g \in \mathcal{C}[a, b]$,

$$(f + g)(x) = f(x) + g(x) = g(x) + f(x) = (g + f)(x).$$

- *Additive identity:* Let 0 denote the zero function $0(x) \equiv 0$. It is continuous and $(f + 0)(x) = f(x) + 0 = f(x)$ for all x , so $f + 0 = f$.

- *Additive inverse:* For $f \in \mathcal{C}[a, b]$, define $-f$ by $(-f)(x) := -f(x)$. Then $-f$ is continuous and $(f + (-f))(x) = f(x) - f(x) = 0$, so $f + (-f) = 0$.

- *Compatibility of scalar multiplication:* For $\alpha, \beta \in \mathbb{C}$ and $f \in \mathcal{C}[a, b]$,

$$((\alpha\beta)f)(x) = (\alpha\beta)f(x) = \alpha(\beta f(x)) = (\alpha(\beta f))(x).$$

- *Identity scalar:* For $f \in \mathcal{C}[a, b]$,

$$(1 \cdot f)(x) = 1 \cdot f(x) = f(x),$$

hence $1 \cdot f = f$.

- *Distributivity over function addition:* For $\alpha \in \mathbb{C}$ and $f, g \in \mathcal{C}[a, b]$,

$$(\alpha(f + g))(x) = \alpha(f(x) + g(x)) = \alpha f(x) + \alpha g(x) = (\alpha f + \alpha g)(x).$$

- *Distributivity over scalar addition:* For $\alpha, \beta \in \mathbb{C}$ and $f \in \mathcal{C}[a, b]$,

$$((\alpha + \beta)f)(x) = (\alpha + \beta)f(x) = \alpha f(x) + \beta f(x) = (\alpha f + \beta f)(x).$$

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All vector space axioms over $\mathbb{C}$ are therefore satisfied, so $\mathcal{C}[a, b]$ is a complex vector space. We first check that $\langle \cdot, \cdot \rangle$ satisfies the axioms of an inner product on a complex vector space.

(1) Conjugate symmetry. For any $u, v \in \mathcal{C}[a, b]$ we have

$$\overline{\langle v, u \rangle} = \overline{\int_a^b \rho(t) v(t) \overline{u(t)} dt} = \int_a^b \rho(t) \overline{v(t)} u(t) dt = \int_a^b \rho(t) u(t) \overline{v(t)} dt = \langle u, v \rangle.$$

(2) Linearity in the first slot. Let $u_1, u_2, v \in \mathcal{C}[a, b]$ and $\alpha, \beta \in \mathbb{C}$. Using linearity of the (Lebesgue or Riemann) integral, we obtain

$$\begin{aligned} \langle \alpha u_1 + \beta u_2, v \rangle &= \int_a^b \rho(t) (\alpha u_1(t) + \beta u_2(t)) \overline{v(t)} dt \\ &= \alpha \int_a^b \rho(t) u_1(t) \overline{v(t)} dt + \beta \int_a^b \rho(t) u_2(t) \overline{v(t)} dt \\ &= \alpha \langle u_1, v \rangle + \beta \langle u_2, v \rangle. \end{aligned}$$

(3) Conjugate linearity in the second slot. This follows from (1) and (2). Indeed, for $u, v_1, v_2 \in \mathcal{C}[a, b]$ and $\alpha, \beta \in \mathbb{C}$,

$$\begin{aligned} \langle u, \alpha v_1 + \beta v_2 \rangle &= \overline{\langle \alpha v_1 + \beta v_2, u \rangle} \\ &= \overline{\alpha \langle v_1, u \rangle + \beta \langle v_2, u \rangle} \\ &= \overline{\alpha} \overline{\langle v_1, u \rangle} + \overline{\beta} \overline{\langle v_2, u \rangle} \\ &= \overline{\alpha} \langle u, v_1 \rangle + \overline{\beta} \langle u, v_2 \rangle. \end{aligned}$$

(4) Positive definiteness. For any $u \in \mathcal{C}[a, b]$,

$$\langle u, u \rangle = \int_a^b \rho(t) |u(t)|^2 dt.$$

Since $\rho(t) > 0$ and $|u(t)|^2 \geq 0$ for all t , the integrand is non-negative, hence $\langle u, u \rangle \geq 0$.

If $\langle u, u \rangle = 0$, then $\int_a^b \rho(t) |u(t)|^2 dt = 0$. Because the integrand is continuous and non-negative and $\rho(t) > 0$, this can happen only if $\rho(t) |u(t)|^2 = 0$ for all $t \in [a, b]$. Thus $|u(t)|^2 = 0$ for all t , so $u(t) = 0$ for all $t \in [a, b]$, that is, $u \equiv 0$. Conversely, if $u \equiv 0$, then clearly $\langle u, u \rangle = 0$. Hence $\langle u, u \rangle = 0$ if and only if $u = 0$.

Combining (1)–(4), we see that $\langle \cdot, \cdot \rangle$ is indeed an inner product on $\mathcal{C}[a, b]$.

Now define

$$\|u\|_2 := \sqrt{\langle u, u \rangle} = \left(\int_a^b \rho(t) |u(t)|^2 dt \right)^{1/2}.$$

On any inner-product space the function $\|u\| := \sqrt{\langle u, u \rangle}$ satisfies the three norm axioms: $\|u\| \geq 0$ and $\|u\| = 0$ iff $u = 0$, $\|\alpha u\| = |\alpha| \|u\|$, and the triangle inequality $\|u + v\| \leq \|u\| + \|v\|$, the last one following from the Cauchy–Schwarz inequality. Therefore $\|\cdot\|_2$ is a norm on $\mathcal{C}[a, b]$.

Hence $\mathcal{C}[a, b]$ equipped with $\langle \cdot, \cdot \rangle$ is an inner-product space over $\mathbb{C}$, and the corresponding norm $\|\cdot\|_2$ makes $\mathcal{C}[a, b]$ into a normed vector space.

II. Chebyshev polynomials of the first kind

(a)

It suffices to show that for $m \neq n$ with $m, n \in \mathbb{N}$,

$$\int_{-1}^1 \frac{T_m(x) T_n(x)}{\sqrt{1-x^2}} dx = 0.$$

Using $T_k(x) = \cos(k \arccos x)$, we have

$$\int_{-1}^1 \frac{T_m(x) T_n(x)}{\sqrt{1-x^2}} dx = \int_{-1}^1 \frac{\cos(m \arccos x) \cos(n \arccos x)}{\sqrt{1-x^2}} dx.$$

Let $t = \arccos x$, so that $x = \cos t$ and $dx = -\sin t dt$. Since $\sqrt{1-x^2} = \sqrt{1-\cos^2 t} = \sin t$ for $t \in [0, \pi]$, we obtain

$$\int_{-1}^1 \frac{\cos(m \arccos x) \cos(n \arccos x)}{\sqrt{1-x^2}} dx = \int_{\pi}^0 \frac{\cos(mt) \cos(nt)}{\sin t} (-\sin t) dt = \int_0^{\pi} \cos(mt) \cos(nt) dt.$$

Now use the identity $\cos a \cos b = \frac{1}{2}(\cos(a+b) + \cos(a-b))$:

$$\int_0^\pi \cos(mt) \cos(nt) dt = \frac{1}{2} \int_0^\pi (\cos((m+n)t) + \cos((m-n)t)) dt.$$

If $m \neq n$, then $m+n \neq 0$ and $m-n \neq 0$, hence

$$\int_0^\pi \cos(kt) dt = \left[\frac{\sin(kt)}{k} \right]_0^\pi = 0 \quad (k \in \mathbb{Z} \setminus \{0\}),$$

so the integral is 0, as desired.

(b)

The first three Chebyshev polynomials of the first are

$$T_0(x) = 1, \quad T_1(x) = x, \quad T_2(x) = 2x^2 - 1.$$

Using the result in (a)

$$\langle T_0, T_0 \rangle = \pi, \quad \langle T_n, T_n \rangle = \frac{\pi}{2} \quad (n \geq 1),$$

we get

$$\|T_0\| = \sqrt{\pi}, \quad \|T_1\| = \sqrt{\frac{\pi}{2}}, \quad \|T_2\| = \sqrt{\frac{\pi}{2}}.$$

Hence an orthonormal system is

$$\phi_0(x) = \frac{T_0(x)}{\|T_0\|} = \frac{1}{\sqrt{\pi}}, \quad \phi_1(x) = \frac{T_1(x)}{\|T_1\|} = \sqrt{\frac{2}{\pi}} x, \quad \phi_2(x) = \frac{T_2(x)}{\|T_2\|} = \sqrt{\frac{2}{\pi}} (2x^2 - 1).$$

Then $\langle \phi_i, \phi_j \rangle = \delta_{ij}$ for $i, j \in \{0, 1, 2\}$ Then is an orthonormal system.

III. Least-squares approximation of $y(x) = \sqrt{1-x^2}$ on $[-1, 1]$

We use the inner product from Theorem 5.7 with

$$\langle f, g \rangle = \int_{-1}^1 f(x)g(x)\rho(x)dx, \quad \rho(x) = \frac{1}{\sqrt{1-x^2}}.$$

Let $f(x) = \sqrt{1-x^2}$. So $\langle f, g \rangle = \int_{-1}^1 g(x)dx$

(a) Fourier expansion

By result in II(b) $T_n(\cos t) = \cos(nt)$.

$$\phi_0(x) = \frac{T_0(x)}{\|T_0\|} = \frac{1}{\sqrt{\pi}}, \quad \phi_1(x) = \frac{T_1(x)}{\|T_1\|} = \sqrt{\frac{2}{\pi}} x, \quad \phi_2(x) = \frac{T_2(x)}{\|T_2\|} = \sqrt{\frac{2}{\pi}} (2x^2 - 1).$$

Compute the coefficients:

$$a_0 = \langle f, \phi_0 \rangle = \int_{-1}^1 \phi_0(x) dx = \int_{-1}^1 \frac{1}{\sqrt{\pi}} dx = \frac{2}{\sqrt{\pi}},$$

$$a_1 = \langle f, \phi_1 \rangle = \int_{-1}^1 \phi_1(x) dx = \int_{-1}^1 \sqrt{\frac{2}{\pi}} x dx = 0,$$

$$a_2 = \langle f, \phi_2 \rangle = \int_{-1}^1 \phi_2(x) dx = \int_{-1}^1 \sqrt{\frac{2}{\pi}} (2x^2 - 1) dx = \sqrt{\frac{2}{\pi}} \left(2 \cdot \frac{2}{3} - 2 \right) = -\frac{2}{3} \sqrt{\frac{2}{\pi}}.$$

Thus

$$p_2(x) = \frac{2}{\sqrt{\pi}} \cdot \frac{1}{\sqrt{\pi}} + \left(-\frac{2}{3} \sqrt{\frac{2}{\pi}} \right) \cdot \sqrt{\frac{2}{\pi}} (2x^2 - 1) = \frac{2}{\pi} - \frac{4}{3\pi} (2x^2 - 1) = \frac{10}{3\pi} - \frac{8}{3\pi} x^2.$$

(b) normal equations

Let $p_2(x) = c_0 + c_1x + c_2x^2$. Then we know

$$\langle f - p_2, 1 \rangle = 0, \quad \langle f - p_2, x \rangle = 0, \quad \langle f - p_2, x^2 \rangle = 0,$$

Since $f(x)$ and $\rho(x) = \frac{1}{\sqrt{1-x^2}}$ are even, while x is odd, we have $\langle f, x \rangle = 0$, $\langle 1, x \rangle = 0$, and $\langle x^2, x \rangle = 0$. Thus $\langle f - p_2, x \rangle = 0$ forces $c_1 = 0$.

Now compute the needed inner products directly:

$$\langle f, 1 \rangle = \int_{-1}^1 1 dx = 2,$$

$$\langle 1, 1 \rangle = \int_{-1}^1 \frac{1}{\sqrt{1-x^2}} dx = \pi, \quad \langle x^2, 1 \rangle = \int_{-1}^1 \frac{x^2}{\sqrt{1-x^2}} dx = \frac{\pi}{2}.$$

Hence $\langle f - p_2, 1 \rangle = 0$ becomes

$$2 - c_0\pi - c_2\frac{\pi}{2} = 0.$$

Then

$$\langle f, x^2 \rangle = \int_{-1}^1 x^2 dx = \frac{2}{3},$$

$$\langle 1, x^2 \rangle = \int_{-1}^1 \frac{x^2}{\sqrt{1-x^2}} dx = \frac{\pi}{2}, \quad \langle x^2, x^2 \rangle = \int_{-1}^1 \frac{x^4}{\sqrt{1-x^2}} dx = \frac{3\pi}{8}.$$

Therefore $\langle f - p_2, x^2 \rangle = 0$ becomes

$$\frac{2}{3} - c_0\frac{\pi}{2} - c_2\frac{3\pi}{8} = 0.$$

Solving the equation

$$c_2 = -\frac{8}{3\pi}, \quad c_0 = \frac{10}{3\pi}, \quad c_1 = 0,$$

hence

$$p_2(x) = \frac{10}{3\pi} - \frac{8}{3\pi}x^2,$$

IV. DLS via orthonormal polynomials

data. We have $N = 12$, $\rho(x) \equiv 1$, and the sales record (Example 5.55) is

$$(x_i, y_i) = (1, 256), (2, 201), (3, 159), (4, 61), (5, 77), (6, 40), (7, 17), (8, 25), (9, 103), (10, 156), (11, 222), (12, 345).$$

Discrete inner product

$$\langle u, v \rangle = \sum_{i=1}^{12} u(x_i)v(x_i).$$

(a)

Then independent list $(1, x, x^2)$. Let $p_0(x) = 1$. Then

$$\langle p_0, p_0 \rangle = \sum_{i=1}^{12} 1 = 12, \quad \phi_0(x) = \frac{p_0(x)}{\|p_0\|} = \frac{1}{\sqrt{12}}.$$

Let $v_1(x) = x$. Remove its projection onto ϕ_0 :

$$\langle v_1, \phi_0 \rangle = \sum_{i=1}^{12} x_i \cdot \frac{1}{\sqrt{12}} = \frac{1}{\sqrt{12}} \sum_{i=1}^{12} i = \frac{78}{\sqrt{12}}.$$

Hence

$$p_1(x) = v_1(x) - \langle v_1, \phi_0 \rangle \phi_0(x) = x - \frac{78}{12} = x - \frac{13}{2}.$$

Moreover,

$$\langle p_1, p_1 \rangle = \sum_{i=1}^{12} \left(i - \frac{13}{2} \right)^2 = 143, \quad \phi_1(x) = \frac{p_1(x)}{\|p_1\|} = \frac{x - \frac{13}{2}}{\sqrt{143}}.$$

Let $v_2(x) = x^2$. Remove its projections onto ϕ_0, ϕ_1 and set

$$p_2(x) = v_2(x) - \langle v_2, \phi_0 \rangle \phi_0(x) - \langle v_2, \phi_1 \rangle \phi_1(x).$$

By computation:

$$p_2(x) = x^2 - 13x + \frac{91}{3}.$$

Furthermore,

$$\langle p_2, p_2 \rangle = \sum_{i=1}^{12} \left(i^2 - 13i + \frac{91}{3} \right)^2 = \frac{4004}{3}.$$

So

$$\phi_2(x) = \frac{3x^2 - 39x + 91}{\sqrt{12012}},$$

Therefore an orthonormal system is

$$\phi_0(x) = \frac{1}{\sqrt{12}}, \quad \phi_1(x) = \frac{x - \frac{13}{2}}{\sqrt{143}}, \quad \phi_2(x) = \frac{3x^2 - 39x + 91}{\sqrt{12012}}.$$

(b) Best quadratic least-squares approximation

We seek $\hat{\varphi}(x) = a_0 + a_1x + a_2x^2$ minimizing

$$\sum_{i=1}^{12} (y_i - \hat{\varphi}(x_i))^2 = \|y - \hat{\varphi}\|^2$$

with respect to the discrete inner product. By (a) use the orthonormal basis. Assume

$$\hat{\varphi}(x) = c_0\phi_0(x) + c_1\phi_1(x) + c_2\phi_2(x), \quad c_k = \langle y, \phi_k \rangle, k = 0, 1, 2$$

Then

$$\begin{aligned} c_0 &= \sum_{i=1}^{12} y_i \cdot \frac{1}{\sqrt{12}} = \frac{\sum_{i=1}^{12} y_i}{\sqrt{12}} = \frac{1662}{\sqrt{12}}, \\ c_1 &= \sum_{i=1}^{12} y_i \cdot \frac{x_i - \frac{13}{2}}{\sqrt{143}} = \frac{\sum_{i=1}^{12} y_i (x_i - \frac{13}{2})}{\sqrt{143}} = \frac{589}{\sqrt{143}}, \\ c_2 &= \sum_{i=1}^{12} y_i \cdot \frac{3x_i^2 - 39x_i + 91}{\sqrt{12012}} = \frac{\sum_{i=1}^{12} y_i (3x_i^2 - 39x_i + 91)}{\sqrt{12012}} = \frac{36204}{\sqrt{12012}}. \end{aligned}$$

Then we get

$$\hat{\varphi}(x) = 386 - \frac{16220}{143}x + \frac{1293}{143}x^2$$

(i.e. $\hat{\varphi}(x) \approx 386 - 113.4265734x + 9.041958042x^2$). which result is same as the result in the notes. More exactly, we can solve the equation in the notes:

$$Ga = c$$

$$G = \begin{pmatrix} 12 & 78 & 650 \\ 78 & 650 & 6084 \\ 650 & 6084 & 60710 \end{pmatrix}, \quad c = \begin{pmatrix} 1662 \\ 11392 \\ 109750 \end{pmatrix},$$

whose solution is exactly

$$a_0 = 386, \quad a_1 = -\frac{16220}{143}, \quad a_2 = \frac{1293}{143},$$

agreeing with the result above.

(c)

If another sales record has the same nodes x_i and N but different values y_i . We suppose the same weight $\rho(x) \equiv 1$ holds. Otherwise both of the methods will compute from start. then:

- The orthonormal polynomials ϕ_0, ϕ_1, ϕ_2 from (a) can be reused, since they depend only on $\{x_i\}$ and ρ .
- Only the coefficients $c_k = \langle y, \phi_k \rangle = \sum_{i=1}^{12} y_i \phi_k(x_i)$ need to be recomputed.

This illustrates an advantage of orthonormal polynomials over normal equations: the orthogonalization basis can be done once, and the new polynomial for new y reduce to computing a few inner products, but normal equation still need to solve a linear equation, which will give orthonormal method better numerical stability.

V. Prove Theorem and lemma

Theorem 5.66

(1)

$$AA^+A = A$$

By the definition of A^+

$$\forall y \in \mathbb{F}^m, A^+y = \sum_{j=1}^r \frac{1}{\sigma_j} \langle y, v_j \rangle u_j$$

Choose y as every column of A , Then we know

$$Ax = y = Ae_j$$

where e_j is the natural o.n.b in $\mathbb{F}^m$, so $\forall z \in \ker(A^*)$

$$\langle y, z \rangle = \langle e_j, A^*z \rangle = 0$$

So $Ax = y$ have solution, which is $\sum_{j=1}^r \frac{1}{\sigma_j} \langle y, v_j \rangle u_j = A^+y$. So $AA^+y = y$ for every column of A . Then combine every column of A together

$$AA^+A = A$$

(2)

$$A^+AA^+ = A^+$$

By definition we know

$$\begin{aligned} \forall y \in \mathbb{F}^m, A^+AA^+y &= A^+A \sum_{j=1}^r \frac{1}{\sigma_j} \langle y, v_j \rangle u_j = A^+ \sum_{j=1}^r \frac{1}{\sigma_j} \langle y, v_j \rangle Au_j = A^+ \sum_{j=1}^r \langle y, v_j \rangle v_j \\ &= \sum_{j=1}^r \langle y, v_j \rangle \sum_{k=1}^r \frac{1}{\sigma_k} \langle v_j, v_k \rangle u_k = \sum_{j=1}^r \langle y, v_j \rangle \frac{1}{\sigma_j} u_j = A^+y \end{aligned}$$

Choose $y = e_j, j = 1, 2, \dots, m$ one by one. Then we know $A^+AA^+ = A^+$

(3)

$$(A^+A)^* = A^+A \quad (AA^+)^* = AA^+$$

$$\forall y, z \in \mathbb{F}^m, AA^+y = \sum_{j=1}^r \langle y, v_j \rangle v_j \Rightarrow \langle AA^+y, z \rangle = \sum_{j=1}^r \langle y, v_j \rangle \langle v_j, z \rangle = \langle y, AA^+z \rangle = \langle y, (AA^+)^*z \rangle$$

Therefore for y, z are arbitrary, so $(AA^+)^* = AA^+$.

$$\forall y, z \in \mathbb{F}^m, A^+Ay = \sum_{j=1}^r \frac{1}{\sigma_j} \langle Ay, v_j \rangle u_j = \sum_{j=1}^r \frac{1}{\sigma_j} \langle y, A^*v_j \rangle u_j = \sum_{j=1}^r \langle y, u_j \rangle u_j$$

$\Rightarrow$

$$\langle A^+Ay, z \rangle = \sum_{j=1}^r \langle y, u_j \rangle \langle u_j, z \rangle = \langle y, A^+Az \rangle$$

Therefore for y, z are arbitrary, so $(A^+A)^* = A^+A$.

Lemma 5.67

Assume $A \in \mathbb{F}^{m \times n}$, When A has linearly independent columns, we know:

$$r(A) = n, \&n \leq m \Rightarrow r(A^*) = r(A) = n$$

Given

$$A^*Ax = 0 \Rightarrow x^*A^*Ax = 0 \Rightarrow Ax = 0$$

Then we can part B from A , s.t. $\exists B \in \mathbb{F}^{n \times n}$ and $|B| \neq 0$, so

$$Bx = 0 \Rightarrow x = 0$$

Therefore $\ker(A^*A) = \{0\}$, then it's invertible. By Thm before

$$(AA^+)^*A = (A^+)^*A^*A = AA^+A = A \Rightarrow A^+ = (A(A^*A)^{-1})^* = (A^*A)^{-1}A^*$$

Then for case A has the linearly independent rows. Take $B = A^*$, then B has linearly independent columns. So B^*B is invertible, therefore AA^* is invertible. Then

$$A(A^+A)^* = AA^*(A^+)^* = AA^+A = A \Rightarrow A^+ = ((A^*A)^{-1}A)^* = A^*(A^*A)^{-1}$$

Which completes the proof.